Solutions Tutorial 9

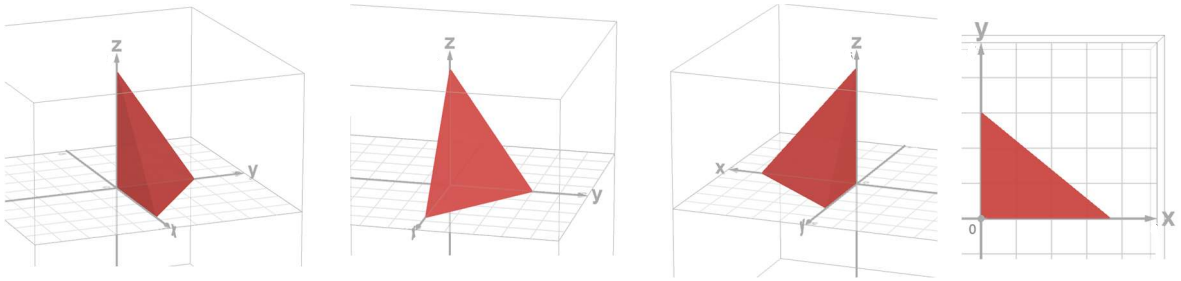
1. Find the volume of the solid region bounded above by the paraboloid $z = 9 - x^2 - y^2$ and bounded below by the unit disc in the xy-plane (i.e., $z = 0$).

Solution. Let $x = r\cos\theta$ and $y = r\sin\theta$. Given $D = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$. Then

$$V = \iint_D (9 - x^2 - y^2)dA = \int_0^{2\pi}\int_0^1 (9 - r^2)r\,dr\,d\theta = \int_0^{2\pi}\left[\tfrac{9}{2}r^2 - \tfrac{1}{4}r^4\right]_0^1 d\theta = \tfrac{17\pi}{2}.$$

2. Find the volume of the tetrahedron bounded by the coordinates planes and the plane whose equation is $ax + by + cz = 1$, where a, b, $c > 0$.

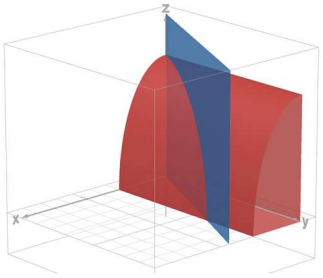
Solution. The plane $ax + by + cz = 1$ cuts the xy-plane in the line $ax + by = 1$:



$$V = \int_0^{1/a}\int_0^{(1-ax)/b} \tfrac{1}{c}(1 - ax - by)dy\,dx = \tfrac{1}{c}\int_0^{1/a}\left[y - axy - \tfrac{1}{2}by^2\right]_0^{(1-ax)/b} dx$$
$$= \tfrac{1}{bc}\int_0^{1/a}\left[(1-ax) - ax(1-ax) - \tfrac{1}{2}(1-ax)^2\right]dx$$
$$= \tfrac{1}{2bc}\int_0^{1/a}(1 - 2ax + a^2x^2)dx = \tfrac{1}{2bc}\left[x - ax^2 + \tfrac{a^2}{3}x^3\right]_0^{1/a} = \tfrac{1}{6abc}$$

3. Find the volume in the first octant (i.e., where x, y, $z \ge 0$) bounded by the cylinder $x^2 + z^2 = 9$ and the plane $y = 2x$.

Solution.



$$V = \int_0^3\int_0^{2x}\sqrt{9 - x^2}\,dy\,dx$$
$$= \int_0^3 2x\sqrt{9 - x^2}\,dx$$
$$= -\left[\tfrac{2}{3}(9 - x^2)^{3/2}\right]_0^3 = 18$$

4. Find $\iiint_R (x + 2y + 4z)\,dV$ where $R = [1,2]\times[-1,0]\times[0,3]$ (is a cuboid).

Solution. $R = \{(x,y,z) : x \in [1,2], y \in [-1,0], z \in [0,3]\}$. Then

$$\iiint_R (x + 2y + 4z)\,dxdydz = \int_0^3\int_{-1}^0\int_1^2 (x + 2y + 4z)\,dxdydz = \tfrac{39}{2} \text{ (verify yourself).}$$

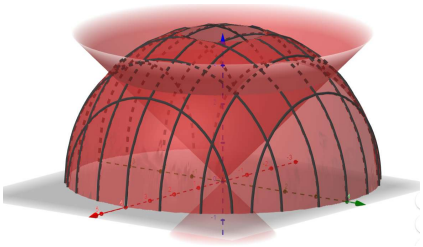
5. Let D be the region bounded by the coordinate planes and the plane with equation $x + y + z = 2$. Find $\iiint_D 1 dxdydz$ and $\iiint_D x dxdydz$.

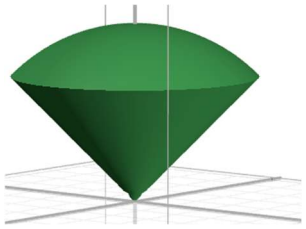
Solution. From Q2, $a = b = c = \frac{1}{2}$. So, $\iiint_D 1 dxdydz = \frac{1}{6(\frac{1}{2})^3} = \frac{4}{3}$. Now, the intervals are $0 \le z \le 2 - x - y$, $0 \le y \le 2 - x$, $0 \le x \le 2$. So,

$$\iiint_D x dxdydz = \int_0^2 \int_0^{2-x} \int_0^{2-y-x} x dzdydx = \frac{1}{2} \int_0^2 x(2-x)^2 dx = \frac{2}{3}.$$

6. Find the volume of the region D bounded by the cone $z^2 = x^2 + y^2$ and the hemisphere $z = \sqrt{16 - x^2 - y^2} > 0$. (Hint: Use cylindrical polar coordinates.)

Solution. Let $x = r\cos\theta$, $y = r\sin\theta$, $z = z$ (unchanged). Then $dxdydz = rdrd\theta dz$.





This is also known as *gasing* in Malay, a traditional game.

Note that $\sqrt{x^2 + y^2} \le z \le \sqrt{16 - x^2 - y^2} \Rightarrow r \le z \le \sqrt{16 - r^2}$.

Solving $r = \sqrt{16 - r^2} \Rightarrow r = \sqrt{8} = 2\sqrt{2}$ so we have $0 \le r \le \sqrt{8} = 2\sqrt{2}$ and $0 \le \theta \le 2\pi$.

$$V = \iiint_D 1 dV = \int_0^{2\pi} \int_0^{2\sqrt{2}} \int_r^{\sqrt{16-r^2}} 1 \, rdzdrd\theta = \int_0^{2\pi} d\theta \int_0^{2\sqrt{2}} (r\sqrt{16 - r^2} - r^2) dr = \frac{64\pi}{3}(2 - \sqrt{2})$$

7. Let D be the region bounded by the surface of the hemisphere $z = \sqrt{1 - x^2 - y^2}$ and the plane $z = 0$. Use spherical polar coordinates to find $\iiint_D z^3 dxdydz$.

Solution. $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta \Rightarrow dxdydz = r^2 \sin\theta drd\theta d\phi$.

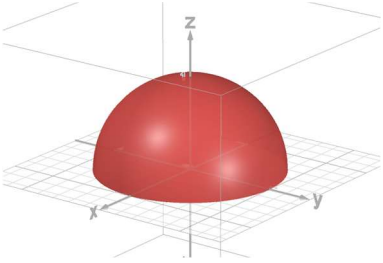


Illustration for D, but the triple integrals for this particular question does not mean the volume, but rather the "mass" of the solid, if f is a density function.

$$\begin{aligned}
\iiint_D z^3 dV &= \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 \left(r^3 \cos^3\theta r^2 \sin\theta\right) drd\theta d\phi \\
&= \frac{1}{6} \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \left[r^6 \cos^3\theta \sin\theta\right]_0^1 d\theta d\phi \\
&= \frac{1}{6} \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} \left(\cos^3\theta \sin\theta\right) d\theta \\
&= -\frac{1}{12}\pi \left[\cos^4\theta\right]_0^{\frac{\pi}{2}} = \frac{1}{12}\pi
\end{aligned}$$

8. Find the volume of the region D bounded by the hemisphere $y=\sqrt{4-x^2-z^2}$ and the planes $y=x$, $y=x\sqrt{3}$ (in the region $x,\ y>0$). [Use spherical polar coordinates]

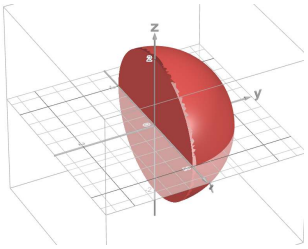
Solution. $x=r\sin\theta\cos\phi,\ y=r\sin\theta\sin\phi,\ z=r\cos\theta \Rightarrow dxdydz=r^2\sin\theta drd\theta d\phi$,

where $0\le\theta\le\pi, \frac{\pi}{4}\le\phi\le\frac{\pi}{3}, 0\le r\le 2$.

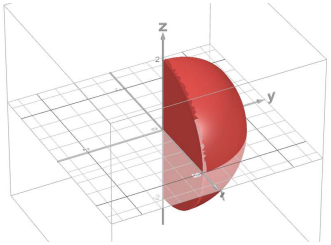
$$V=\int_0^{\pi}\int_{\pi/4}^{\pi/3}\int_0^2 1\times r^2\sin\theta\, drd\phi d\theta=\int_0^{\pi}\sin\theta d\theta\int_0^2 r^2dr\int_{\pi/4}^{\pi/3}d\phi=2(\tfrac{8}{3})(\tfrac{\pi}{3}-\tfrac{\pi}{4})=\tfrac{4}{9}\pi$$

Sketch the graph:

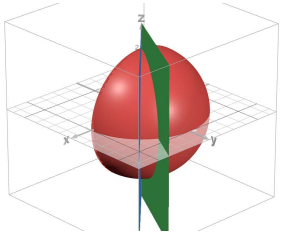
(i) Sketch $y=\sqrt{4-x^2-z^2}$.



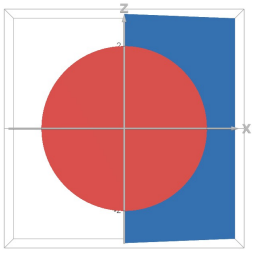
(ii) Add the constraints $x,\ y>0$.



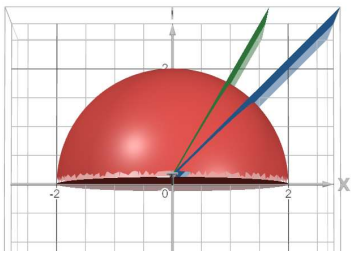
(iii) Draw two planes: $y=x, y=x\sqrt{3}$.



(iv) For θ, view the xz-plane: $0\le\theta\le\pi$.
For r, view the xz-plane: $0\le r\le 2$.



(v) For ϕ, view the xy – plane: $\frac{\pi}{4}\le\phi\le\frac{\pi}{3}$.



Remark: Note that for this question, we may interchange y and z.
$y=\sqrt{4-x^2-z^2}$ to $z=\sqrt{4-x^2-y^2}$;
and $0<y=x,\ y=x\sqrt{3}$ to
$0<z=x, z=x\sqrt{3}$.

9. Find the volume of the ellipsoid which is given by $\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2+\left(\frac{z}{c}\right)^2\le 1\,,\ a,b,c>0\,.$

Solution. We use the given hint: $x=ar\sin\theta\cos\phi, y=br\sin\theta\sin\phi, z=cr\cos\theta\,.$

$$\begin{aligned}
J &= \det\begin{pmatrix} a\sin\theta\cos\phi & ar\cos\theta\cos\phi & -ar\sin\theta\sin\phi \\ b\sin\theta\sin\phi & br\cos\theta\sin\phi & br\sin\theta\cos\phi \\ c\cos\theta & -cr\sin\theta & 0 \end{pmatrix} \\
&= abcr^2\sin\theta\begin{vmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{vmatrix} \\
&= abcr^2\sin\theta\left(\cos\theta\begin{vmatrix} \cos\theta\cos\phi & -\sin\phi \\ \cos\theta\sin\phi & \cos\phi \end{vmatrix}+\sin\theta\begin{vmatrix} \sin\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\phi \end{vmatrix}\right) \\
&= abcr^2\sin\theta\Big(\cos^2\theta+\sin^2\theta\Big)=abcr^2\sin\theta
\end{aligned}$$

Since $\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2+\left(\frac{z}{c}\right)^2\le 1$, it follows that

$$\begin{cases} \frac{x}{a}=r\sin\theta\cos\phi \\ \frac{y}{b}=r\sin\theta\sin\phi \\ \frac{z}{c}=r\cos\theta \end{cases}\Rightarrow r^2\sin^2\theta(\cos^2\phi+\sin^2\phi)+r^2\cos^2\theta=r^2\le 1\,,$$

so r runs from 0 to 1.

We take the first octant (so that $J>0$) and multiply the answer by 8, the volume is

$$8\int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\int_0^1\Big(abcr^2\sin\theta\Big)drd\phi d\theta=8abc\int_0^{\frac{\pi}{2}}\sin\theta d\theta\int_0^1 r^2dr\int_0^{\frac{\pi}{2}}d\phi=8abc(1\times\tfrac{1}{3}\times\tfrac{\pi}{2})=\tfrac{4}{3}\pi abc\,.$$

Extra exercises

10. Find the total mass of the rectangular box $[0,1]\times[0,2]\times[0,3]$ (this means that $0\le x\le 1\,, 0\le y\le 2\,,\ 0\le z\le 3\,$), with density function $h(x,y,z)=x\,$.

Solution. Formula for mass with density function $\rho(x,y,z)$ over solid region R is

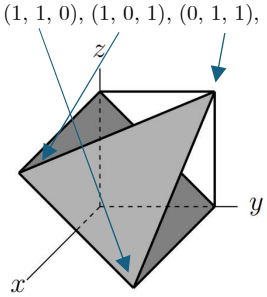
$$\iiint_R \rho(x,y,z)\ dV\,.$$

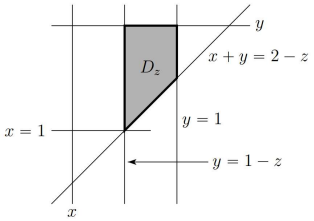
So the mass of the box $[0,1]\times[0,2]\times[0,3]$ with density function $h(x,y,z)=x$ is

$$\iiint_R x\ dV=\int_0^1 x\ dx\int_0^2 dy\int_0^3 dz=\tfrac{1}{2}(2\times 3)=3\,.$$

11. Evaluate $\iiint_R y\ dV$ where R is the portion of the cube $0\le x,y,z\le 1$ lying above the plane $y+z=1$ and below the plane $x+y+z=2\,$.

Solution. $R=\Big\{(x,y,z):0\le x\le 2-y-z,\ 1-z\le y\le 1,\ 0\le z\le 1\Big\}$.





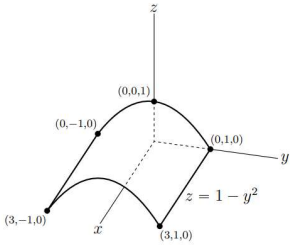
$$\iiint_R y\ dV=\int_0^1\int_{1-z}^1\int_0^{2-y-z} y\ dxdydz$$
$$=\int_0^1\int_{1-z}^1 y(2-y-z)\ dydz$$
$$=\int_0^1\Big[y^2-\tfrac{y^3}{3}-\tfrac{1}{2}zy^2\Big]_{1-z}^1\ dz \qquad \boxed{\tfrac{1}{2}z-z^2+\tfrac{1}{2}z^3}$$
$$=\int_0^1\Big(1-\tfrac{1}{3}-\tfrac{1}{2}z\Big)-\Big((1-z)^2-\tfrac{(1-z)^3}{3}-\tfrac{1}{2}z(1-z)^2\Big)\ dz$$
$$=\Big[\tfrac{2}{3}z-\tfrac{1}{4}z^2+\tfrac{1}{3}(1-z)^3-\tfrac{1}{12}(1-z)^4+\tfrac{1}{4}z^2-\tfrac{1}{3}z^3+\tfrac{1}{8}z^4\Big]_0^1$$
$$=\tfrac{2}{3}-\tfrac{1}{4}+\tfrac{1}{4}-\tfrac{1}{3}+\tfrac{1}{8}-\tfrac{1}{3}+\tfrac{1}{12}=\tfrac{5}{24}$$

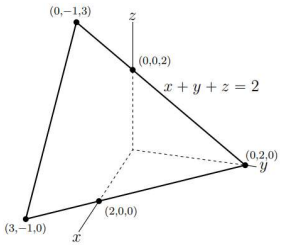
12. A triple integral $\iiint_E f\,dV$ is given $\displaystyle\int_{y=-1}^{y=1}\int_{z=0}^{z=1-y^2}\int_{x=0}^{x=2-y-z} f(x,y,z)\ dxdzdy\,.$

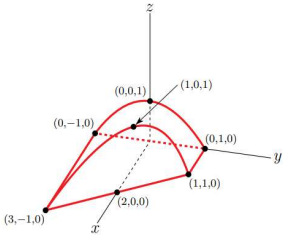
(a) Draw E in 3-dimensions. Be sure to show the units on the coordinate axes.

(b) Rewrite $\iiint_E f\,dV$ in the order of $dzdxdy$.

Solution. (a)







(b) Note that $x+y+z=2$ and $z=1-y^2$, so $x+y+1-y^2=2 \Rightarrow x=y^2-y+1$.

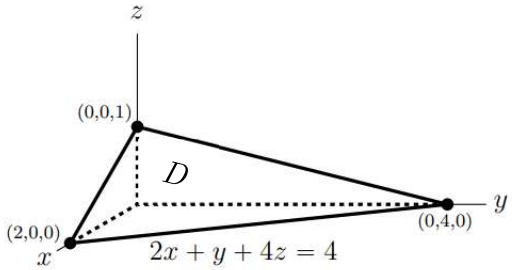
$$\int_{y=-1}^{y=1}\int_{z=0}^{z=1-y^2}\int_{x=0}^{x=2-y-z} f(x,y,z)\,dxdzdy = \int_{y=-1}^{y=1}\int_{x=0}^{x=1+y^2-y}\int_{z=0}^{z=1-y^2} f(x,y,z)\,dzdxdy + \int_{y=-1}^{y=1}\int_{x=1+y^2-y}^{x=2-y}\int_{z=0}^{z=2-x-y} f(x,y,z)\,dzdxdy$$

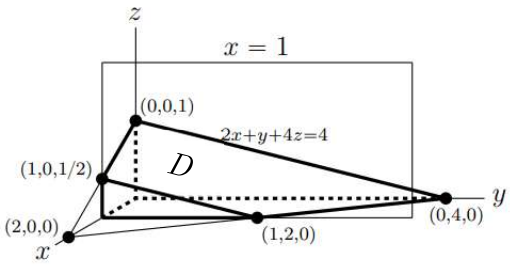
13. A triple integral $\iiint_E f\,dV$ is given $J=\int_0^1\int_0^{1-\frac{x}{2}}\int_0^{4-2x-4z} f(x,y,z)\,dydzdx$.

(a) Sketch the domain E in 3-dimensions. Be sure to show the units.

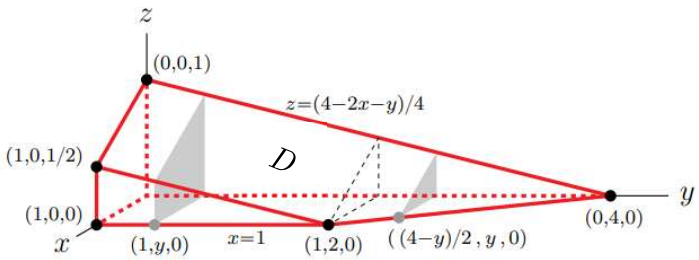
(b) Rewrite $\iiint_E f\,dV$ in the order of $dzdxdy$.

Solution. (a) Note that $2x+y+4z=4$ is plane D.





$0 \le y \le 4-2x-4z$ & $0 \le z \le 1-\frac{x}{2}$ and with $0 \le x \le 1$

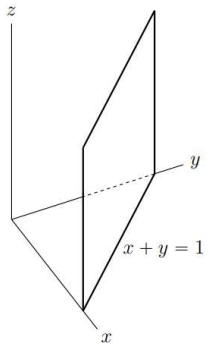


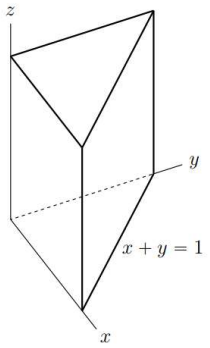
(b) Intersection line of plane D $(2x+y+4z=4)$ and xy–plane $(z=0)$, is $2x+y=4$ or $x=2-\frac{y}{2}$.

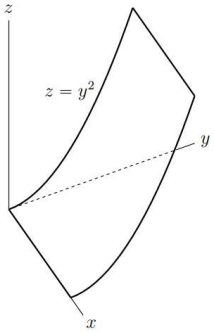
$$J=\int_{y=0}^{y=2}\int_{x=0}^{x=1}\int_{z=0}^{z=1-\frac{x}{2}-\frac{y}{4}} f(x,y,z)\,dzdxdy + \int_{y=2}^{y=4}\int_{x=0}^{x=2-\frac{y}{2}}\int_{z=0}^{z=1-\frac{x}{2}-\frac{y}{4}} f(x,y,z)\,dzdxdy$$

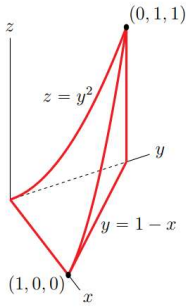
14. Let E be the region in the first octant bounded by the coordinate planes, the plane $x+y=1$ and the surface $z=y^2$. Evaluate $\iiint_E z\,dV$.

Solution.









$$\iiint_E z\,dV=\int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=y^2} z\,dzdydx=\tfrac{1}{2}\int_{x=0}^{x=1}\int_{y=0}^{y=1-x} y^4dydx=\tfrac{1}{10}\int_{x=0}^{x=1}(1-x)^5dydx=\tfrac{1}{60}$$

15. Evaluate $\iiint_R yz^2e^{-xyz}\,dV$ over $R=\{(x,y,z):0\le x\le 1,\ 0\le y\le 2,\ 0\le z\le 3\}$.

Solution. $\displaystyle\int_{z=0}^{z=3}\int_{y=0}^{y=2}\int_{x=0}^{x=1} yz^2e^{-xyz}\,dxdydz=\int_{z=0}^{z=3}\int_{y=0}^{y=2}\left[-ze^{-xyz}\right]_0^1 dydz=\int_{z=0}^{z=3}\int_{y=0}^{y=2}\left(-ze^{-yz}+z\right)dydz$

$\displaystyle=\int_{z=0}^{z=3}\left[e^{-yz}+zy\right]_0^2 dz=\int_{z=0}^{z=3}\left(e^{-2z}+2z-1\right)dz=\left[-\tfrac{1}{2}e^{-2z}+z^2-z\right]_0^3=\tfrac{1}{2}(13-e^{-6})$

16. Convert from cylindrical to Cartesian coordinates:

(a) $r=1,\ \theta=0,\ z=0$ (b) $r=1,\ \theta=\frac{\pi}{4},\ z=0$

Solution. (a) $\left(x,y,z\right)=(r\cos\theta,r\sin\theta,z)=\left(1,0,0\right)$ (b) $\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0\right)$.

17. Convert from Cartesian to cylindrical coordinates: (a) (1, 1, 2) (b) (–1, –1, 2)

Solution. (a) $z=2,\ r^2=x^2+y^2=2,\ 1=\tan\theta\Rightarrow\theta=\frac{\pi}{4}+2n\pi, n\in\mathbb{Z}$

(b) $z=0,\ r^2=2\Rightarrow r=\sqrt{2},\ \tan\theta=1\Rightarrow\theta=\frac{\pi}{4}+2n\pi, n\in\mathbb{Z}$

18. Rewrite the following equations in cylindrical coordinates:

(a) $z=2xy$ (b) $x^2+y^2+z^2=1$ (c) $(x-1)^2+y^2=1$

Solution. (a) $z=2xy=2r^2\cos\theta\sin\theta=r^2\sin(2\theta)$.

(b) $x^2+y^2+z^2=1\Rightarrow r^2+z^2=1$.

(c) $(x-1)^2+y^2=1\Rightarrow(r\cos\theta-1)^2+r^2\sin^2\theta=1\Rightarrow r^2=2r\cos\theta$.

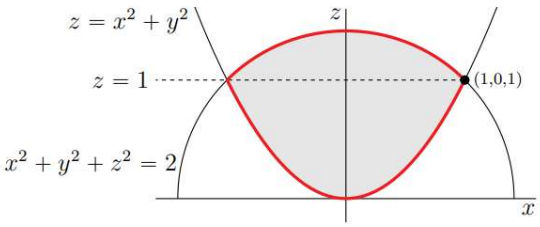
So $r=2\cos\theta$ when $r>0$.

19. Let E be the region bounded between the parabolic surfaces $z = x^2 + y^2$ and $z = 2 - x^2 - y^2$ and within the cylinder $x^2 + y^2 \le 1$. Calculate the integral of $f(x,y,z) = (x^2 + y^2)^{3/2}$ over the region E.

Solution. $\iiint_E f\,dV = \int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=1}\int_{z=r^2}^{z=2-r^2} r^3\; rdzdrd\theta = 2\pi\int_{r=0}^{r=1} r^4(2-2r^2)\;dr = \frac{8\pi}{35}$.

20. Let E be the region bounded above by the sphere $x^2 + y^2 + z^2 = 2$ and below by the paraboloid $z = x^2 + y^2$. Find the centroid of E.

Solution. Suppose E is invariant under rotations around the z–axis. Then the centroid (centre of mass) of E will lie on the z–axis. Thus $\overline{x} = 0 = \overline{y}$ and $\overline{z} = \dfrac{\iiint_E z dV}{\iiint_E dV}$. Now $z = x^2 + y^2 = r^2$, $x^2 + y^2 + z^2 = 2 \Rightarrow z^2 = 2 - r^2$. The cross section of E when $y = 0$:



We evaluate $\int_0^1\int_0^{2\pi}\int_{r^2}^{\sqrt{2-r^2}} z^n r dz d\theta dr$ for $n = 0$ and 1.

$$\int_0^1\int_0^{2\pi}\int_{r^2}^{\sqrt{2-r^2}} z^n r dz d\theta dr = \int_0^1\int_0^{2\pi}\left[\frac{z^{n+1}}{n+1}\right]_{r^2}^{\sqrt{2-r^2}} r d\theta dr$$

$$= \begin{cases} 2\pi\int_0^1 \left(r\sqrt{2-r^2} - r^3\right) dr, n = 0 \\ \frac{2\pi}{2}\int_0^1 \left(2r - r^3 - r^5\right) dr, \quad n = 1 \end{cases}$$

$$= \begin{cases} 2\pi\left[-\frac{1}{3}\left(2-r^2\right)^{3/2} - \frac{1}{4}r^4\right]_0^1, n = 0 \\ \pi\left[r^2 - \frac{1}{4}r^4 - \frac{1}{6}r^6\right]_0^1, \qquad n = 1 \end{cases}$$

$$= \begin{cases} -2\pi\left(\frac{1}{3} + \frac{1}{4} - \frac{2}{3}\sqrt{2}\right), n = 0 \\ \pi\left(1 - \frac{1}{4} - \frac{1}{6}\right), \qquad n = 1 \end{cases}$$

$$= \begin{cases} -2\pi\left(\frac{7}{12} - \frac{2}{3}\sqrt{2}\right), n = 0 \\ \frac{7}{12}\pi, \qquad n = 1 \end{cases}$$

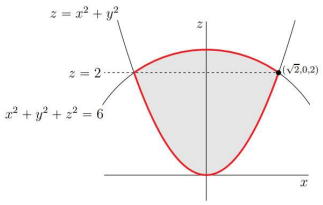
So, $\overline{z} = \frac{\frac{7}{12}}{\frac{4}{3}\sqrt{2} - \frac{7}{6}} = \frac{7}{16\sqrt{2}-14}$ and the centroid of E is $\left(0, 0, \frac{7}{16\sqrt{2}-14}\right)$.

21. Let E be the smaller of the two solid regions bounded by the surfaces $z = x^2 + y^2$ and $x^2 + y^2 + z^2 = 6$. Evaluate $\iiint_E (x^2 + y^2)\, dV$.

Solution. Note that $z = x^2 + y^2 = r^2$, $x^2 + y^2 + z^2 = 6 \Rightarrow z^2 = 6 - r^2$. The cross section of E when $y = 0$:

$$\int_0^{\sqrt{2}} \int_0^{2\pi} \int_{r^2}^{\sqrt{6-r^2}} r^2 r\, dz\, d\theta\, dr = 2\pi \int_0^{\sqrt{2}} \left[r^3 z \right]_{r^2}^{\sqrt{6-r^2}} dr$$

$$= 2\pi \int_0^{\sqrt{2}} \left(r^3 \sqrt{6 - r^2} - r^5 \right) dr$$



Note that $\int_0^{\sqrt{2}} r^5\, dr = \frac{8}{6} = \frac{4}{3}$. Let $u = 6 - r^2 \Rightarrow du = -2r dr$. Then

$$\int_0^{\sqrt{2}} \left(r^3 \sqrt{6 - r^2} \right) dr = -\frac{1}{2} \int_4^6 \left(6u^{1/2} - u^{3/2} \right) du = \frac{1}{2} \left[4u^{3/2} - \frac{2}{5} u^{5/2} \right]_4^6$$

$$= \frac{1}{2} \left(4\sqrt{6^3} - \frac{2}{5}\sqrt{6^5} - 4\sqrt{4^3} + \frac{2}{5}\sqrt{4^5} \right) = 12\sqrt{6} - \frac{36}{5}\sqrt{6} - 16 + \frac{32}{5} = \frac{24}{5}\sqrt{6} - \frac{48}{5}$$

Hence, $\int_0^{\sqrt{2}} \int_0^{2\pi} \int_{r^2}^{\sqrt{6-r^2}} r^2 r\, dz\, d\theta\, dr = 2\pi \left(\frac{24}{5}\sqrt{6} - \frac{48}{5} - \frac{4}{3} \right) = 2\pi \left(\frac{24}{5}\sqrt{6} - \frac{164}{15} \right)$.

22. Let E be the solid lying above the surface $z = y^2$ and below the surface $z = 4 - x^2$. Evaluate $\iiint_E y^2\, dV$.

Solution. Note that $y^2 \le z \le 4 - x^2 \Rightarrow r^2 \sin^2\theta \le z \le 4 - r^2 \cos^2\theta$.

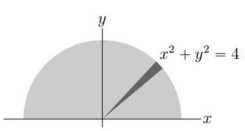
$$\int_0^2 \int_0^{2\pi} \int_{r^2 \sin^2\theta}^{4 - r^2\cos^2\theta} r^2 \sin^2\theta\, r\, dz\, d\theta\, dr = \int_0^2 \int_0^{2\pi} r^3 \sin^2\theta \left(4 - r^2\cos^2\theta - r^2 \sin^2\theta \right) d\theta\, dr$$

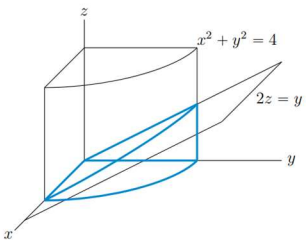
$$= \int_0^2 \int_0^{2\pi} r^3 \sin^2\theta \left(4 - r^2 \right) d\theta\, dr = \frac{1}{2} \int_0^2 \int_0^{2\pi} r^3 \left(4 - r^2 \right) \left(1 - \cos 2\theta \right) d\theta\, dr \quad \because 1 - 2\sin^2\theta = \cos 2\theta$$

$$= \frac{1}{2} \int_0^2 \left[r^3 \left(4 - r^2 \right) \left(\theta - \frac{1}{2}\sin 2\theta \right) \right]_0^{2\pi} dr = \pi \int_0^2 r^3 \left(4 - r^2 \right) dr = \pi \left[r^4 - \frac{1}{6} r^6 \right]_0^2 = \frac{16}{3}\pi$$

23. Find the volume of a solid inside $x^2 + y^2 = 4$, above $z = 0$ and below $2z = y$.

Solution. Note that $x^2 + y^2 = 4 \Rightarrow r = 2$ and $z = \frac{1}{2} y = \frac{r}{2} \sin\theta$.





$$\int_0^{\pi} \int_0^2 \int_0^{\frac{1}{2} r \sin\theta} r\, dz\, dr\, d\theta = \int_0^{\pi} \int_0^2 r \left(\frac{r \sin\theta}{2} \right) dr\, d\theta$$

$$= \frac{8}{6} \int_0^{\pi} \sin\theta\, d\theta = -\frac{4}{3} (\cos\pi - \cos 0) = \frac{8}{3}$$

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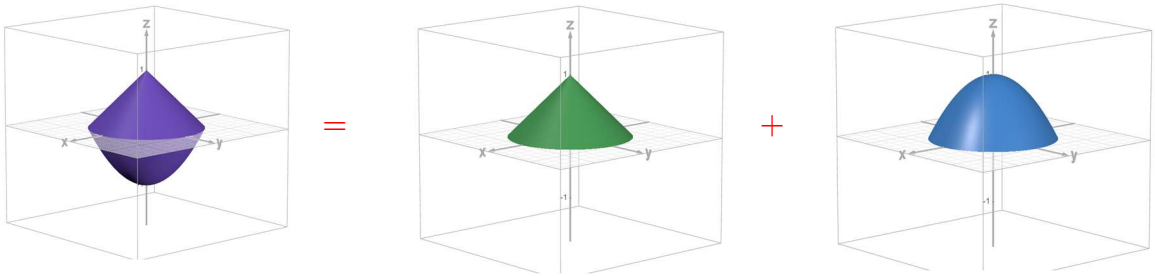
Summary

Volume	Mass	Centroid
$\iiint_D 1 \; dxdydz$	$\iiint_R f(x,y,z) \; dxdydz$ f is the density function.	E is invariant under rotations around the z–axis. Then the centroid (centre of mass) of E will lie on the z–axis. Thus $\overline{x} = 0 = \overline{y}$ and $\overline{z} = \dfrac{\iiint_E z \; dV}{\iiint_E dV}$.

Past test/final exam questions

24. Determine the volume of the 3-dimensional region D inside the cylinder $x^2+y^2=1$, bounded above by the surface $z=1-\sqrt{x^2+y^2}$ and below by the surface $z=x^2+y^2-1$.

Solution. Note that we can compute the volume of the cone in the first octant; and compute the volume of the paraboloid in the first octant after we rewrite the function as $z=1-x^2-y^2$.

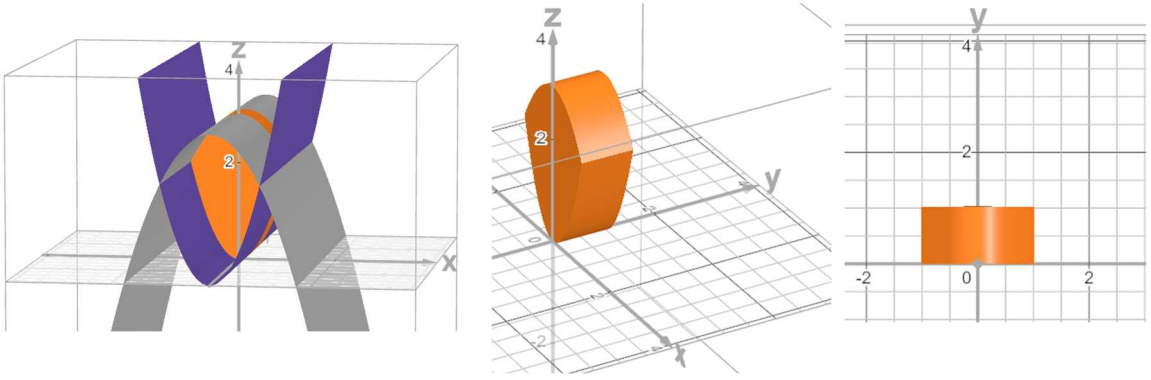


$$V=4\int_0^{\frac{\pi}{2}}\int_0^1\Big((1-r)+(1-r^2)\Big)rdrd\theta=4\times\tfrac{\pi}{2}\Big[r^2-\tfrac{r^3}{3}-\tfrac{r^4}{4}\Big]_0^1=2\pi(1-\tfrac{1}{3}-\tfrac{1}{4})=\tfrac{5}{6}\pi$$

Alternative: $4\int_0^{\frac{\pi}{2}}\int_0^1\int_{x^2+y^2-1}^{1-\sqrt{x^2+y^2}} rdrd\theta=4\int_0^1\int_0^{\frac{\pi}{2}}\Big((1-r)-(r^2-1)\Big)rd\theta dr=\ldots=\tfrac{5}{6}\pi$.

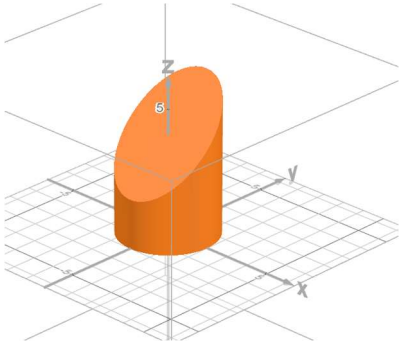
25. Use a triple integral to calculate the volume of the solid which is bounded by $z=3-x^2$, $z=2x^2$, $y=0$, and $y=1$.

Solution. $V=\int_0^1\int_0^1 V=2\int_{2x^2}^{3-x^2}1dzdydx=2\int_0^1\int_0^1(3-3x^2)dydx=6\Big[x-\tfrac{x^3}{3}\Big]_0^1=4$.



26. Use a triple integral to calculate the volume of the solid which is bounded by $z = y + 4$, $z = 0$, and $x^2 + y^2 = 4$.

Solution.



$$V = \int_0^{2\pi}\int_0^{2}\int_0^{y+4} r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^{2}(y+4)r\,dr\,d\theta = \int_0^{2\pi}\int_0^{2}(r\sin\theta+4)r\,dr\,d\theta$$
$$= \int_0^{2\pi}\left[\tfrac{r^3}{3}\sin\theta + 2r^2\right]_0^2 d\theta = 8\int_0^{2\pi}\left(\tfrac{1}{3}\sin\theta + 1\right)d\theta = 8\left[\theta - \tfrac{1}{3}\cos\theta\right]_0^{2\pi} = 16\pi.$$

27. Find the volume of a sphere with radius $a > 0$. Hence, write the volume of a hemisphere with radius a. Use the spherical polar coordinate substitution.

Proof. Let $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$. Then

$$J = \det\begin{pmatrix} \sin\theta\cos\phi & r\cos\theta\cos\theta & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix} = r^2\sin\theta$$

So $dxdydz = r^2\sin\theta dr d\theta d\phi$ for $0 \le \theta, \phi \le \frac{\pi}{2}, 0 \le r \le a$. Hence,

$$V_{\text{sphere}} = 8\int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\int_0^{a} r^2\sin\theta dr d\phi d\theta = 8\int_0^{\frac{\pi}{2}}\sin\theta d\theta\int_0^{a} r^2 dr\int_0^{\frac{\pi}{2}} d\phi = 8(\tfrac{a^3}{3})(\tfrac{\pi}{2}) = \tfrac{4}{3}a^3\pi,$$

and $V_{\text{hemisphere}} = \frac{2}{3}a^3\pi$.